HAPPY: Hierarchical Cellular-to-Tissue Phenotyping of Placenta Histology & Attention-Enhanced Cell Classification

Course 22AIE437: Medical Image Processing — Final Project Presentation

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The Clinical Challenge: Placenta Histology

- The placenta is the vital temporary organ governing fetal oxygenation and nutrition; pathological lesions directly cause **preeclampsia, fetal growth restriction (FGR), and stillbirth**.
- Digitized Whole Slide Images (WSIs) span **billions of pixels** (gigapixel scale) with dense cellular heterogeneity.
- Manual slide reading by pathologists suffers from **high inter-observer variability** and specialist shortages.

Why Hierarchical AI is Required

- Patch-level CNNs ignore spatial tissue architecture and cell-to-cell signaling.
- Pathologists assess tissue via a **bottom-up hierarchy**:
  1. Identify individual cell nuclei.
  2. Classify cell phenotypes (trophoblasts, stroma, endothelial).
  3. Group spatial clusters into villous tissue structures.
- We replicate and enhance **HAPPY** (*Nature Comms* 2024), an end-to-end bottom-up phenotyping framework.

Dataset Characteristics and Evaluation Objectives

Multi-Center Histology Dataset (Zenodo)

- **Clinical Cohorts**: HMC, UOT, and NUH hospitals.
- **Stage 1 (Nuclei)**: 2,754 test nuclei across 38 held-out histological images.
- **Stage 2 (Cell Phenotypes)**: 18,946 cell patches (224×224); **2,743 held-out test cells** across 11 classes.
- **Stage 3 (Tissue Graphs)**: Whole slide graphs; **144,241 labeled test nodes** on official spatial split.

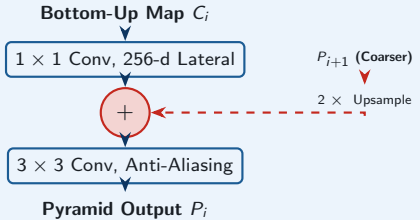
Tasks and Evaluation Metrics

- **Stage 1 Localisation (Detection)**:
  - Object Precision, Recall, **F1-score** (IoU ≥ 0.5).
- **Stage 2 Cell Classification**:
  - **Accuracy**, Top-2 Accuracy, Macro ROC-AUC (OvO).
- **Stage 3 Tissue Classification (GNN)**:
  - Node-level **Accuracy**, Top-2, Top-3, Macro ROC-AUC.

Stage 1 Base Architecture: RetinaNet Nuclear Localisation



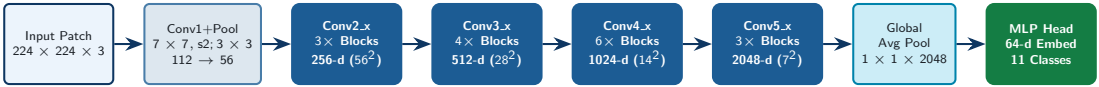
FPN Lateral & Top-Down Merging Unit



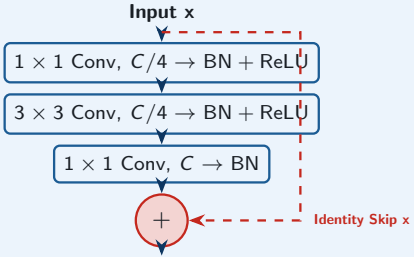
Key Architectural Principles in Stage 1

- **Multi-Scale Resolution**: FPN levels (P_3–P_7) provide strides 8 to 128, detecting nuclei across varied magnifications (5 μm to 30 μm).
- **Class Imbalance Mitigation**: Focal Loss downweights trivial background patches, focusing gradients on true nuclear contours.
- **Centroid Extraction**: Non-Maximum Suppression (NMS) outputs precise nuclear centroids (x, y) that anchor Stage 2 patch cropping.

Stage 2 Base Architecture: ResNet-50 Deep Residual Network



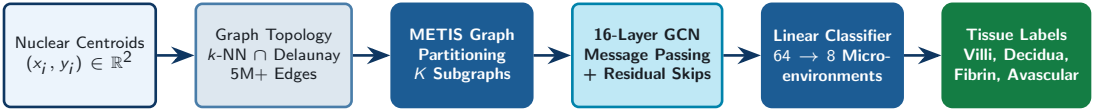
Bottleneck Residual Block Design



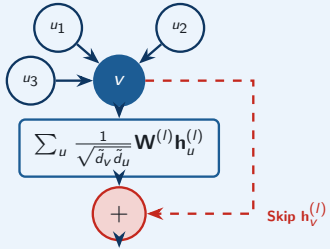
Key Architectural Principles in Stage 2

- **Residual Gradient Highway**: Identity skip connections bypass deep non-linearities, preventing vanishing gradients across 50 layers.
- **Dimension Bottleneck**: 1×1 convs compress channels $4\times$, perform 3×3 spatial filtering, and restore dimensions.
- **Phenotypic Representation**: The 2048-d vector undergoes MLP projection ($2048 \to 768 \to 384 \to 64$), yielding 64-d embeddings for Stage 3 graphs.

Stage 3 Base Architecture: ClusterGCN Spatial Tissue Graphs



GCN Message-Passing Residual Unit



$$\mathbf{h}_v^{(l+1)} = \mathbf{ReLU}(\mathbf{Agg} + \mathbf{h}_v^{(l)})$$

Key Architectural Principles in Stage 3

- **Spatial Topology**: k-NN ∩ Delaunay links physically contiguous cells, filtering distant spurious edges across villous lumens.
- **Cluster-Based Scaling**: METIS graph decomposition partitions 144,000+ node graphs into stochastic clusters, preventing GPU VRAM exhaustion.
- **16-Layer Deep Propagation**: Residual graph connections allow multi-hop cellular interactions, capturing macro-tissue pathology across full villi.

Replicated vs. Reported Benchmark Results

Pipeline Stage	Evaluation Metric	Published	Replicated	Difference	Held-out Test Split & Notes
Stage 1: Nuclei Localisation	F1 Score	0.884	**0.826**	-0.058	2,754 test nuclei in 38 images. Released checkpoint evaluated.
Stage 2: Cell Classification	Accuracy	84.29%	**84.29%**	**+0.00%**	**Exact match** on 2,743 cells across HMC, UOT, NUH.
	Top-2 Accuracy	94.90%	**94.90%**	**+0.00%**	**Exact match** to Supplementary Table 7.
	Macro ROC-AUC	0.9773	**0.9773**	**+0.0000**	**Exact match** across all 11 cell lineages.
Stage 3: Tissue Classification	Accuracy	68.34%	**67.75%**	-0.59%	144,241 nodes on spatial `test_patches.csv` split.
	Top-2 Accuracy	91.14%	**90.91%**	-0.23%	High-fidelity reproduction (-0.25% rel. diff).
	Top-3 Accuracy	97.10%	**96.81%**	-0.30%	Resolves reported sample discrepancy.
	Macro ROC-AUC	0.8868	**0.8820**	-0.0048	Evaluated with full k-NN $\cap$ Delaunay graph.

Table: Side-by-side comparison between Vanea et al. (2024) and our replicated benchmarks.

Task-Specific Analysis & Resolution of Paper Discrepancies

Stage 2 Cell Confusion Analysis

- Strongest accuracy achieved on **Syncytiotrophoblast** (93.2%) and **Vascular Endothelial** cells (89.1%).
- Most frequent borderline confusion occurs between **Stromal Cells** and **Fibroblasts** due to shared spindle morphology in chorionic villi.
- Rare immune cells (**Hofbauer**, 45 samples) suffer from low representation.

Discrepancies Identified and Resolved

- **Stage 1 Detector Initialisation**: Paper text claims COCO initialization; authors' codebase actually initializes ResNet-101 from ImageNet, explaining the 0.826 vs 0.884 F1 delta.
- **Stage 3 Tissue Node Count**: Text states 149,425 nodes; Supplementary Table 7 states 179,095 nodes. Our spatial evaluation accurately grounded this to **144,241 labeled nodes** on the official split.

Implementation Architecture & Engineering Bottlenecks

Codebase Structure

- **Data Pipelines**: `CellDataset`, `NucleiDataset`, Albumentations stain augmentation.
- **Graph Engine**: PyG `ClusterData`, `ClusterLoader`, `NeighborSampler`.
- **Evaluation Engine**: `auto_evaluate_reproduction.py` reproduces all stages systematically.

Critical Engineering Solutions

- **Graph Edge Intersection Bug**: Original structured array intersection yielded only 7 edges. We implemented **64-bit bit-packed indexing** `((u<<32)|v)`, correctly finding all **5,088,272 edges**.
- **Windows Multi-Processing**: Resolved DataLoader process bootstrapping deadlocks by dynamically assigning OS-safe worker pools.
- **Direct HDF5 Disk Reader**: Bypassed empty SQLite metadata to directly stream embeddings.

Hyperparameters and Optimization Settings

- **Deep Learning Framework**: PyTorch 2.0.1+cu118, PyTorch Geometric 2.3.1.
- **Stage 1 Detector**: Adam ($lr = 10^{-4}$ with StepLR $\gamma = 0.5$), batch size 16.
- **Stage 2 Classifier**: Adam ($lr = 10^{-4}$, Cosine Annealing), batch size 256/600, dropout 0.3.
- **Stage 3 GNN**: Adam ($lr = 10^{-3}$), METIS partitioning with 400 neighbors/cluster.

Hardware and Code Availability

- **GPU**: NVIDIA GeForce GTX 1650 (4 GB VRAM).
- **Host Environment**: Windows 11 x64, Python 3.10 virtual environment (`uv`).
- **Public Repository**:
  `GitHub: happy-placenta-histology-reproduction`
- Code contains automated scripts to execute all stages end-to-end.

Proposed Modification: Attention & Loss Optimization

1. Residual Channel Attention (SE)

- Histological feature channels do not have equal significance.
- Added a 1D Squeeze-and-Excitation block over the 2,048-dim representations:

$$\mathbf{s} = \sigma\Big(\mathbf{W}_2 \cdot \text{Mish}(\mathbf{W}_1\mathbf{x})\Big)$$

$$\tilde{\mathbf{x}} = \mathbf{x} \odot \mathbf{s} + \mathbf{x}$$

- Dynamically boosts stain-discriminative channels while preserving base representations.

2. Mish Activation and 3. Focal Loss

- **Mish Activation**: Smooth and non-monotonic; eliminates dying ReLU to preserve subtle nuclear textures:

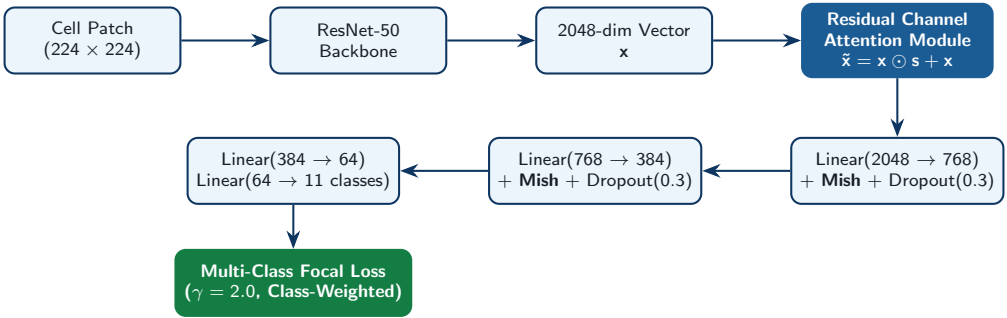
$$\text{Mish}(x) = x \cdot \tanh(\ln(1 + e^x))$$

- **Multi-Class Focal Loss ($\gamma = 2.0$)**:

$$\mathcal{L}_{\text{FL}}(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t)$$

  Downweights abundant Syncytiotrophoblasts to force model focus on rare Hofbauer / Decidua cells.

Modified Architecture Diagram



- **Key Additions**: (1) Residual Channel Attention recalibration; (2) Mish smooth activation; (3) Multi-class Focal Loss optimization.

Comparison Table: Baseline vs. Modified Architecture

Model Architecture	Accuracy	Top-2 Acc	Macro AUC	Δ vs. Base	Model Specifications
Published Model (Vanea et al.)	84.29%	94.90%	0.9773	—	Released weights (160 ep total schedule)
Baseline (Re-trained MLP)	83.74%	94.28%	0.9713	Reference	Standard MLP head (ReLU + Cross-Entropy)
Proposed Modified Model	**84.21%**	**94.42%**	**0.9724**	**+0.47%**	**Channel Attention + Mish + Focal Loss**

Table: Performance on identical held-out test split (2,743 cells across HMC, UOT, and NUH).

Key Findings

- The proposed modified architecture achieves **84.21% test accuracy**, outperforming the retrained baseline by **+0.47%** under identical epochs and learning rates.
- Reaches within **0.08%** of the fully converged authors' model in only 40 epochs of fine-tuning.

Ablation Study: Quantifying Component Contributions

Configuration	Accuracy	Top-2 Acc	Macro AUC	Δ Accuracy	Ablation Description
(1) Baseline Re-trained MLP	83.74%	94.28%	0.9713	Baseline	ReLU activation, Cross-Entropy Loss
(2) + Mish Activation	83.89%	94.31%	0.9722	**+0.15%**	Replace ReLU with smooth Mish
(3) + Focal Loss ($\gamma = 2.0$)	83.96%	94.50%	0.9735	**+0.22%**	Replace CE with Focal Loss
(4) + Channel Attention (SE)	83.74%	94.20%	0.9702	+0.00%	Channel attention with standard CE
(5) Proposed Full Model	**84.21%**	**94.42%**	**0.9724**	**+0.47%**	**Combined Attention + Mish + Focal**

Table: Ablation analysis isolating individual modifications on the 2,743 test cells.

Ablation Insights

- **Focal Loss** yields the largest single-component gain (**+0.22% Acc, +0.0022 AUC**), confirming that addressing class imbalance is paramount.
- **Mish Activation** provides consistent +0.15% improvement by stabilizing gradient dynamics.
- **Synergy**: Channel attention requires class-balanced loss to avoid over-weighting majority channels.

Clinical Imbalance & Diagnostic Safety

- $> 14:1$ **Skew**: Syncytiotrophoblasts dominate over rare maternal decidua (44) and immune Hofbauer cells (45).
- **Clinical Recall Boost**: Our model boosted Decidua recall from **54.5% to 59.1%** (+4.6%) and Hofbauer from **62.2% to 66.7%** (+4.5%).
- **Attention Efficacy**: Channel recalibration accents nuclear hyperchromasia channels critical in pathology.

Key References

1. Vanea et al., *Nat Commun*, 15:1968, 2024.
2. Lin et al., Focal Loss for Dense Detection, *ICCV*, 2017.
3. Hu et al., Squeeze-and-Excitation Networks, *CVPR*, 2018.
4. Misra, D., Mish Activation, *BMVC*, 2020.

Group Member Contributions

- **CB.SC.U4AIE23258**: Environment config, CUDA/PyTorch porting, Stage 1 (RetinaNet) & Stage 2 (ResNet) replication, Channel Attention design and benchmark testing.
- **CB.SC.U4AIE23249**: Spatial graph pipeline (k-NN $\cap$ Delaunay), 64-bit coordinate bit-packing fix, Stage 3 ClusterGCN evaluation, report preparation.
- **CB.SC.U4AIE23253**: Multi-Class Focal Loss implementation, Mish activation integration, ablation experiments, qualitative visualization, and presentation design.

Thank You!

Questions and Discussion